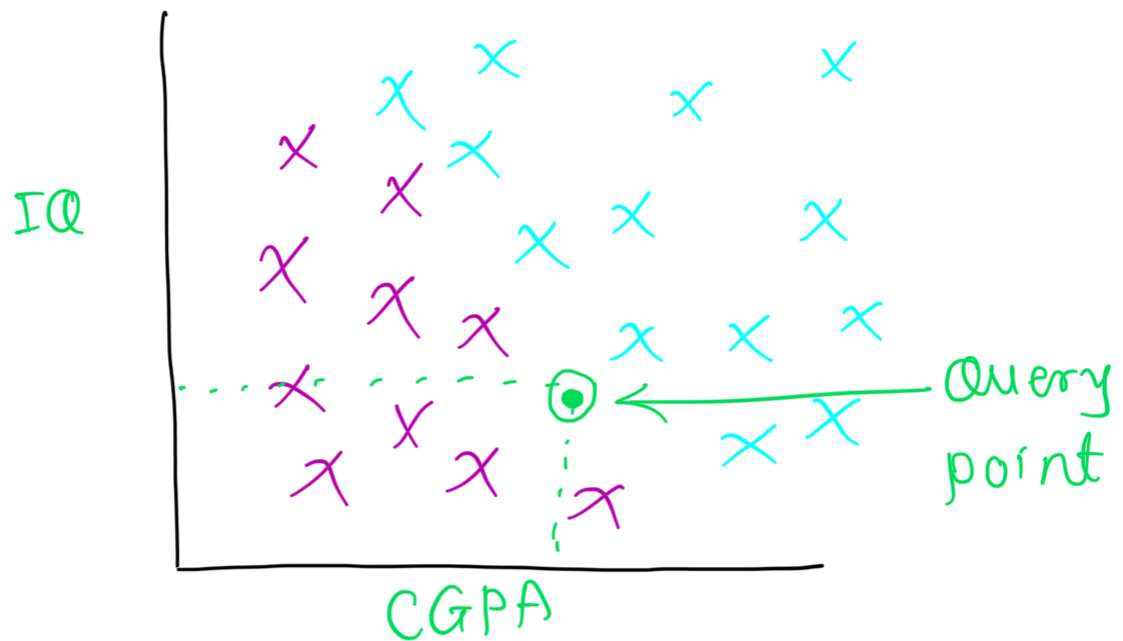


K - Nearest Neighbour

$K = 3 \Rightarrow 3$ nearest neighbours



Majority Count is applied on most nearest 3 point (as $K=3$) & then class label of the query point

K - Nearest Neighbour (KNN) is a supervised learning method that classifies new data points based on the

proximity & majority class of their nearest neighbours in existing dataset

It is a "lazy learner" because it stores all the training data & performs computations only when a prediction is needed, rather than building a general model.

How KNN works: step by step

1) Select the number of neighbours: (k)

A user defined constant is chosen, determining how many neighbouring data points the algorithm will consider.

2) Calculate distance:

The algorithm calculates the distance between the new data point & every single point in the training dataset using a relevant distance metric, such as Euclidean distance, Manhattan distance or Hamming distance.

3) Identify nearest neighbours.

The calculated distances are sorted & the top k data points with the smallest distances are selected

4) Assign the class C for classification

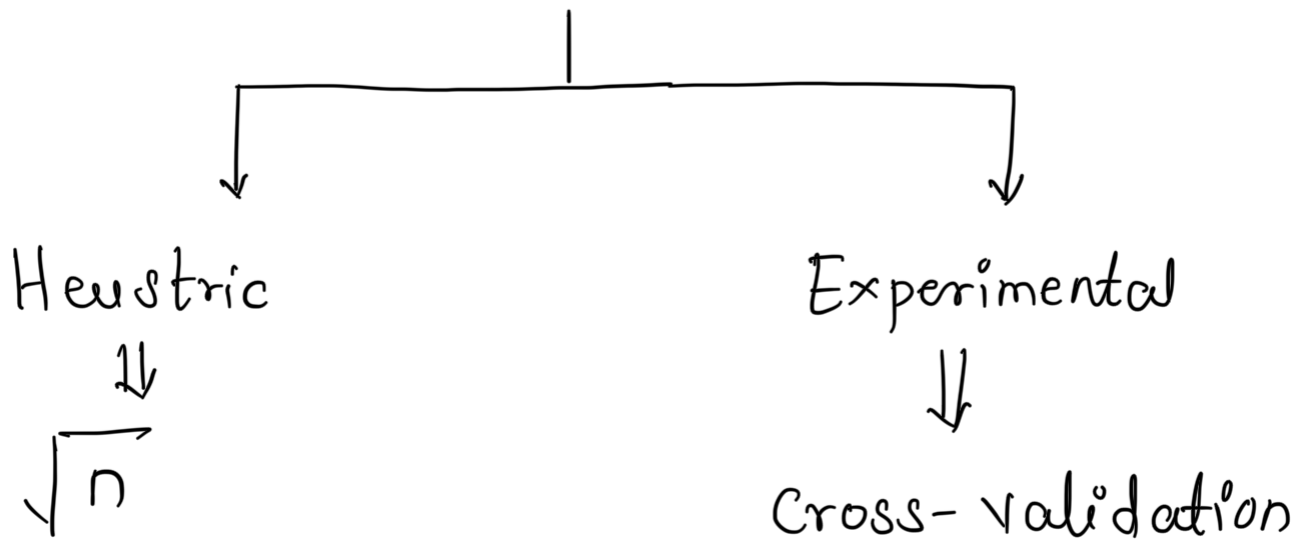
The algorithm counts the number of data points in each category among these k neighbours. The new data point is then assigned the category that

appears most frequently (a majority vote)

5) Predict the value (for regression)

If used for regression task, the algorithm calculates the average (mean) of the target value of the k nearest neighbours and this average becomes the predicted value for new point.

How To select k ?



n = observations

+ or - To make value
of k odd